Fiscal Year:	FY 2009	**Task Last Updated:**	FY 04/04/2010
PI Name:	Hlatky, Lynn Ph.D.		
Project Title:	Computational Modeling of Chromosome Aberrations Produced by HZE Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:	PI affiliation name change to Steward Research and Specialty Projects Corporation (2012). PI formerly at Caritas St. Elizabeth's Medical Center, Boston, MA (2005-2011); Dana-Farber Cancer Institute, Boston, MA (prior to 2005). Subsequent name change from Steward Research and Specialty Projects Corporation to GeneSys Research Institute in 2013.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	05/15/2004	**End Date:**	09/30/2010
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: grant # changed in 2006 when PI moved--new number is NNJ06HA27G; also, end date is now 9/30/2010, per S. Krenek/JSC (3/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hahnfeldt, Philip (Caritas St. Elizabeth's Medical Center, Tufts Univ School of Medicine) Sachs, Rainer (University of California)		
Grant/Contract No.:	NNJ06HA27G ; NNJ04HF42G		
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Task Description:	Chromosome aberrations produced by high LET radiations are directly and indirectly relevant to space exploration and risk estimation in a number of different ways. They are important as probes for DNA repair and recombination mechanisms, as biomarkers of dose and radiation quality, as factors in genomic instability and carcinogenesis, and as surrogates for cancer in risk estimation. Quantitative modeling is needed to exploit and augment the rapidly accumulating empirical data on aberration types and frequencies. This project will apply mathematical and computer algorithms to data on aberrations produced by iron ions (56Fe). The analysis will be compared to analyses of damage produced by photons, electrons, or alpha particles, with the goal of assessing signature responses to 56Fe, an important radiation encountered in space. To perform such studies, the team has tied damage endpoints to risk through our customized software and algorithms. These endpoints include: 1. aberration complexity, quantified by cycle structures; 2. apparently and truly incomplete aberrations, related to segment size distributions; 3. aberration spectra as related to chromosome geometry, to DNA double strand break (DSB) clustering depending on radiation track structure, and especially to DSB misrejoining depending on proximity; 4. mFISH color junctions for each specific chromosome pair, related to chromosome-chromosome spatial correlations in an interphase nucleus. Substantial preliminary studies have recently been carried out by our team for each aim. As remarked, we will employ the team's specialized and comprehensive software, including the highly-developed chromosome aberration simulator (CAS) software, for analyzing the aberration spectra encountered. The data to be analyzed will be for human cells in vitro, scored by mFISH or other similar techniques. A number of these data sets are being made available by many of our colleagues worldwide. This will greatly expand the range of suitable data available for analysis. We have also recently devised methods for systematically comparing mFISH data with data obtained by older methods such as FISH or solid staining and with newer methods such as color banding. For Specific Aim 1, we will replace the current approach of using "obligate" (deterministic) cycles with a more accurate computer-based approach involving cycle probabilities. For Aim 2, new methods are being developed that allow systematic comparison of apparently incomplete aberrations with corresponding complete aberrations and corresponding truly incomplete ones, even in the case of very complex aberrations. Under Aim 2 we will also consider size distributions of fragments that have one telomere and one unrejoined free end, for comparison to recent data using pan-telomeric probes; and analyze mixtures of different radiations. Aim 3 will emphasize more realistic computer algorithms for break misrejoining, bringing this aspect of aberration modeling more nearly in line with the very sophisticated models now being used by various groups for assessing the influence of track structure and chromatin geometry on DSB locations. For Aim 4 we plan to apply a method we previously used for low LET data to data for high LET, including iron ions. The method involves testing frequencies of color junction pairs for deviations from randomness.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Improved biodosimetry
Task Progress:	Two main projects were carried out. 1. Rapid alteration approximation for cell populations undergoing chromosomal instability. We continued the mathematical analysis of aberrations which appear during carcinogenesis and cancer progression, discussed in the previous progress report. We emphasized the chromosomal instability that can arise from HZE radiation, and can arise from cell-cell fusion or other horizontal transfer processes as recently observed using 24 color chromosome painting in our lab. Such instabilty involves chromosome aberrations and other large-scale alterations of the karyotype. Alterations that change the fitness of a cell and its progeny during carcinogenesis and cancer progression are often modeled using population dynamics equations for a diverse somatic cell population. Usually it has been assumed, paralleling standard mathematical approaches to evolution, that such alterations are slow compared to fitness selection, i.e. compared to cell subpopulation frequency changes induced by unequal subpopulation net proliferation rates. However, in agreement with the literature we found chromosomal instability alterations can be rapid, with major karyotype changes within a few passages in vitro. Rapid alterations call for starting from an approximation to cell proliferation and alteration processes different from the slow-alteration approximations usually considered. We therefore developed a new approach to modeling these processes. We initiated a mathematical study of situations where alterations are rapid compared to selection. To this end we introduced a proliferation-alteration system of ordinary differential equations which generalizes selection-mutation systems suggested in the literature earlier. Our proliferation-alteration system has the following three properties, which have not previously, as far as we are aware, been combined in a cell population dynamics approach: * The equations involve cell subpopulation numbers not just cell subpopulation frequencies. * Rather than just tracking independent subpopulations, the equations incorporate cell-cell interactions of a general type, where each subpopulation net proliferation rate (i.e. the instantaneous fitness of that subpopulation) can depend, in general non-linearly, on all the cell subpopulation numbers. Cell-cell interactions are currently considered important in radiation-induced carcinogenesis and cancer progression. In contrast to typical replicator-mutator equations (reviewed, e.g., in Komarova NL, 2005, Mathematical modeling of tumorigenesis: mission possible, Curr Opin Oncol 17:39-43) our equations have alteration terms which are explicitly separated out from the proliferation terms and are comparatively simple, allowing the implementation of a "rapid-alteration" approximation procedure. Under surprisingly general circumstances a system-theoretical estimate of the total-population net growth rate (positive or negative) emerged from analyzing the proliferation-alteration equations. This rate characterizes the diverse, interacting cell population acting as a single system; it is a weighted average of subpopulation rates. The intuitive picture is that when alterations are rapid the population, though diverse, tends to grow or decrease as a unit. For example if one cell subpopulation has a comparatively large net proliferation rate the result is not that the frequency of that subpopulation in the total population increases markedly. Rather, that subpopulation also loses many cells through alterations; the altered cells increase the cell numbers of other cell subpopulations, and thus the high fitness of one cell

subpopulation contributes mainly to the overall growth of the entire population rather than just the growth of the highly fit subpopulation. When the population grows as a unit, its overall Malthusian parameter is the average fitness. By comparing with numerical integrations we found that our perturbation approach is highly accurate during periods of very rapid fitness altering cell alterations. The implication is that that during such periods cancer should be modeled as a diverse "ecological" system of interacting cells, rather than as one increasingly malignant clone. For example, suppose that one non-invasive subpopulation proliferates rapidly, that another subpopulation is slow growing but invasive, and that the first can alter rapidly into the second. Then the two subpopulations grow in step, and it is their interrelation rather than the individual properties of either which is the key to carcinogenesis. This scenario of interacting populations gives a possible mechanism for maintaining the clonal heterogeneity that is typical of late (progression) stages of solid tumors. Cancers, and cancer cell lines in vitro, are typically aneuploid, with highly heterogeneous populations, and with individual cell lineages observed to undergo rapid chromosome copy number alterations. However, various authors have commented on the surprising temporal stability (called "stability within instability" by Gusev et al., 2001, Long-term dynamics of chromosomal instability in cancer: a transition probability model, Mathematical and Computer Modelling 33:1253-1273) of average total chromosome number, given the heterogeneity of the population. It could be that this stability within instability is the result of rapid alterations that determine a stable subpopulation frequency distribution even while the overall cell number is increasing, as in our mathematical analysis.

To summarize, we devised a population dynamical approach for analyzing an interacting, diverse, proliferating, rapidly altering somatic cell population of the kind that occurs in chromosomal instability. The approach is plausibly applicable to restricted time intervals during various carcinogenesis and cancer progression processes. It predicts that the overall growth rate is an emergent parameter of the entire interacting tumor cell system, given by an appropriate average, rather than a property of any one clone.

2. Further investigation of the LQE equation.

The classic linear-quadratic-exponential (LQE) equation for radiogenic cancer initiation was originally obtained by analyzing the effect of chromosome aberrations and of other genome alterations on cell initiation and killing. We continued our analysis of the LQE formalism combined with longer-time-scale models in situations of prolonged irradiation, where cell killing can lead to cell repopulation. We used data on incidence of iatrogenic second cancers after tumor radiotherapy, taking into account cell population dynamics for the heterogeneous cell population during and after fractionated protocols and stochastic effects implied by the fact that in early stages the number of pre-malignant cells can be very small, to compare the relative effects of cell initiation and promotion in radiation carcinogenesis.

We found that radiation-induced cancer promotion is particularly important in older individuals, where it can dominate initiation and can lead to a leveling-off or even increase of excess relative risk with increasing time after exposure. These results should hold for radiation carcinogenesis in general not just for second cancers and thus be important in risk estimation.

Bibliography Type:	Description: (Last Updated: 11/01/2023)
Articles in Peer-reviewed Journals	Shuryak I, Hahnfeldt P, Hlatky L, Sachs RK, Brenner DJ. "A new view of radiation-induced cancer: integrating short- and long-term processes. Part I: approach." Radiat Environ Biophys. 2009 Aug;48(3):263-74. PubMed PMID: 19536557 , Aug-2009
Articles in Peer-reviewed Journals	Shuryak I, Hahnfeldt P, Hlatky L, Sachs RK, Brenner DJ. "A new view of radiation-induced cancer: integrating short- and long-term processes. Part II: second cancer risk estimation." Radiat Environ Biophys. 2009 Aug;48(3):275-86. PubMed PMID: 19499238 , Aug-2009